Security Threat And Attack

Cyber Security

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What is security?

-The state of being free from danger or threat is security.

-Controls that are put in place to ensure the confidentiality, security, and availability of all computer system components are known as computer security.

-These components include data, software, hardware, and firmware.

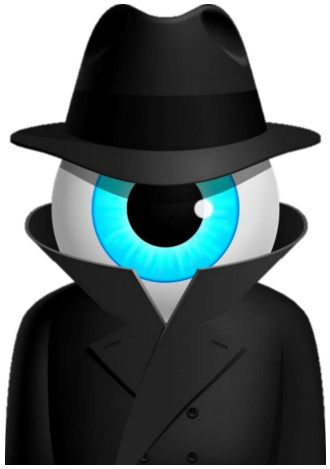
-The word "cyber" denotes a relationship with information technology (IT), i.e., computers.

-Cybersecurity is a procedure created to safeguard devices and networks from outside Attacks.

SECURITY THREAT AND ATTACK

Does threat and attack mean the same?
->NO

THREAT	ATTACK
Can be intentional or Unintentional.	Intentional
May or may not be malicious	Is Malicious
Ability to cause damage	Objective to cause damage
Hard to detect	Easy to detect
Can be blocked by control of vulnerabilities	Cannot be blocked by controlling of vulnerabilities
Can be classified into physical threat, internal threat, external threat, human threat,and non-physical thread.	Can be classified into virus, spyware, phishing, worms, spams etc.



Security Threat:

-Threats to computer security is a risk to the effectiveness and performance of your computer.

-A computer system threat is a potential risk that could endanger the security of your data.

Types of Security Threat:

1.Physical Threat

2.Non-physical Threat

Security Threats

Physical Threats

Internal

-Short circuit,fire, Non stable power supply,hardware failure etc

External

-Natural Disasters

Human

-Destruction, theft, robbery

Non-Physical Threats

-Malware
-Virus
-Worms
-Trojan

-Hampering data, hacking ID and password, keeping track upon others computer illegally

Some of the types of famous cyber security threats

1.Malware

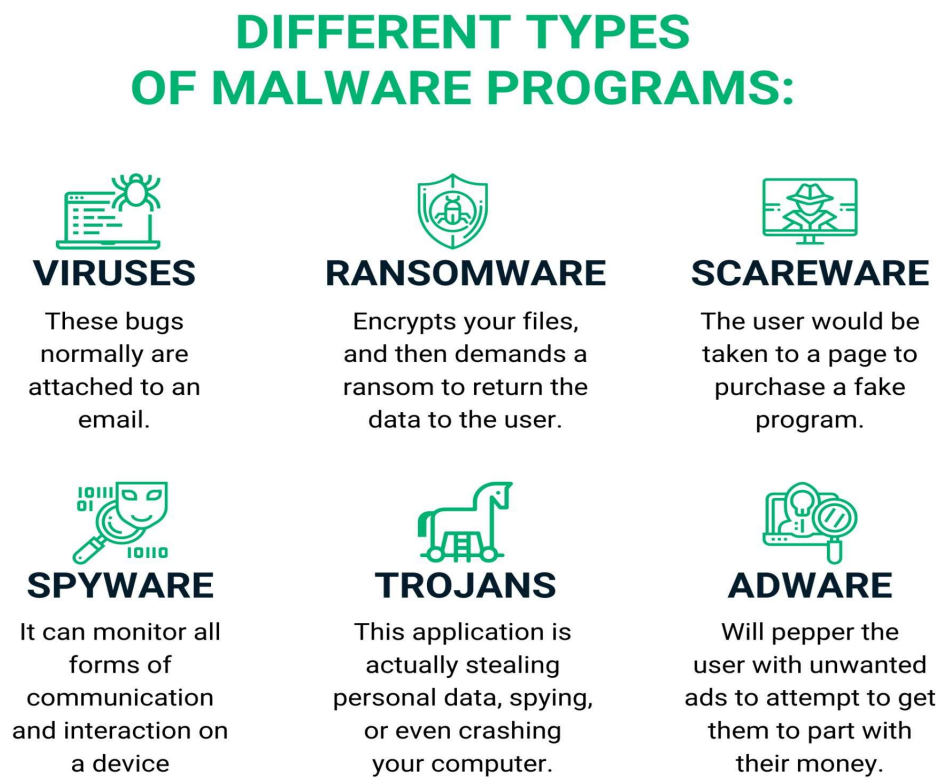
2.Phishing

3.Man in the Middle Attack

4.Denial Of Service Attack

5.Ransomware

6.DNS Attack



Source:sectigo.com

1) Malware

The most frequent dangers to cyber security are malware assaults. When a user follows a risky link or email, harmful software such as spyware, ransomware, viruses, and worms are loaded into the system and are referred to as malware. Malware can damage the system, collect sensitive data, and more after it has gained access to it. It can also block access to crucial network components.

2) Phishing

Malicious emails are sent by cybercriminals that appear to be from reliable sources. The victim is then duped(tricked) into clicking the email's malicious link, which results in the installation of malware or the revealing of private data like login passwords and credit card information.

3) Man in the Middle Attack

An attack known as a "Man in the Middle" (MitM) occurs when online criminals insert themselves in the middle of a two-party conversation. Once the attacker has understood the conversation, they may filter and steal sensitive information while providing the user with various responses.

4) Denial of Service Attack

Attacks known as denial of service attempt to overload servers, networks, or systems with a large volume of traffic, preventing them from responding to valid requests. The target system may potentially be attacked via a number of infected devices. A distributed denial of service (DDoS) assault is what this is.

5)Ransomware

In a malware attack known as ransomware, the victim's data is locked or encrypted, and the attacker makes a threat to either disclose the data or limit access to it until a ransom is paid. Companies can better prevent and respond to ransomware threats by becoming more knowledgeable about them.

6) DNS Attack

A cyberattack known as a DNS assault takes advantage of holes in the Domain Name System (DNS). The attackers use DNS vulnerabilities to extract data from compromised computers and direct site users to malicious pages (DNS Hijacking) (DNS Tunneling)

Let’s discuss about the Effect of

Cyber Threat And Attack

It is categorized into three groups:

1.primary consequences.

2.secondary consequences. and

3.second-order consequences.

1.Primary effects

The main effects are those that certain gadgets within IT systems have directly on users. To produce a cyber event that will disrupt and/or exploit a targeted device, threat actors use a number of steps. This can take many different forms, from aiming to just compromise the data to having a more catastrophic impact like ransomware.

2.Secondary effects

The indirect repercussions brought on by a cyber event are known as secondary effects. When intended, these impacts result from the initial assault on the intended targets' devices.This, again, can be broken down into three areas: time, cost, and reputation.

2.a.Time

Cyberattacks cause businesses to lose time while they wait for IT systems to be protected, recover data that may have been lost in the intrusion, and finally put the systems back online. When discussing squandered time, it does unintentionally result in financial loss.

2.b.Cost

The majority of businesses will experience financial loss as a result of a cyber-attack, whether it be through sales lost as a result of production delays, money taken by cybercriminals, or costs associated with hiring experts to stop the attack.

2.c.Reputation

A good client reputation might take years to establish, but all it takes is one significant cyber attack to destroy the trust. Depending on the scale of the company, it might receive media attention or other companies might profit from it.

3.Second-order effects

The indirect repercussions of this disturbance on society are known as second-order effects. The organization targeted and the end-to-end supply chain they are a part of determine the intensity of these attacks.

Reason that causes security threat and attack

-Clicking on suspicious links

-Not installing Antivirus Softwares

-Not keeping system updated

-Keeping password easy to crack

-Using cracked version softwares

How can we be safe from Security Threats and attacks?

-Install firewall software and keep it updated every time.

-Use passwords that are strong and challenging to guess, including large and small letters, digits, and unusual characters.

-Install antivirus and anti-spyware, and make sure it's always updated.

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-Frequently scan your entire system.

-Verify if installing an application is safe before doing so (using Antivirus Software).

-Read emails that contain attachments with great caution.

Some additional points:

-Limit Sensitive Personal Info on Social Media

-Enable Privacy and Security Settings

-Use a Password Manager

-Limit Social Logins

-Beware of Public Wi-Fi

-Use a VPN (virtual private network) service.

-Use the latest Web browsers because they have improved security for fake websites.

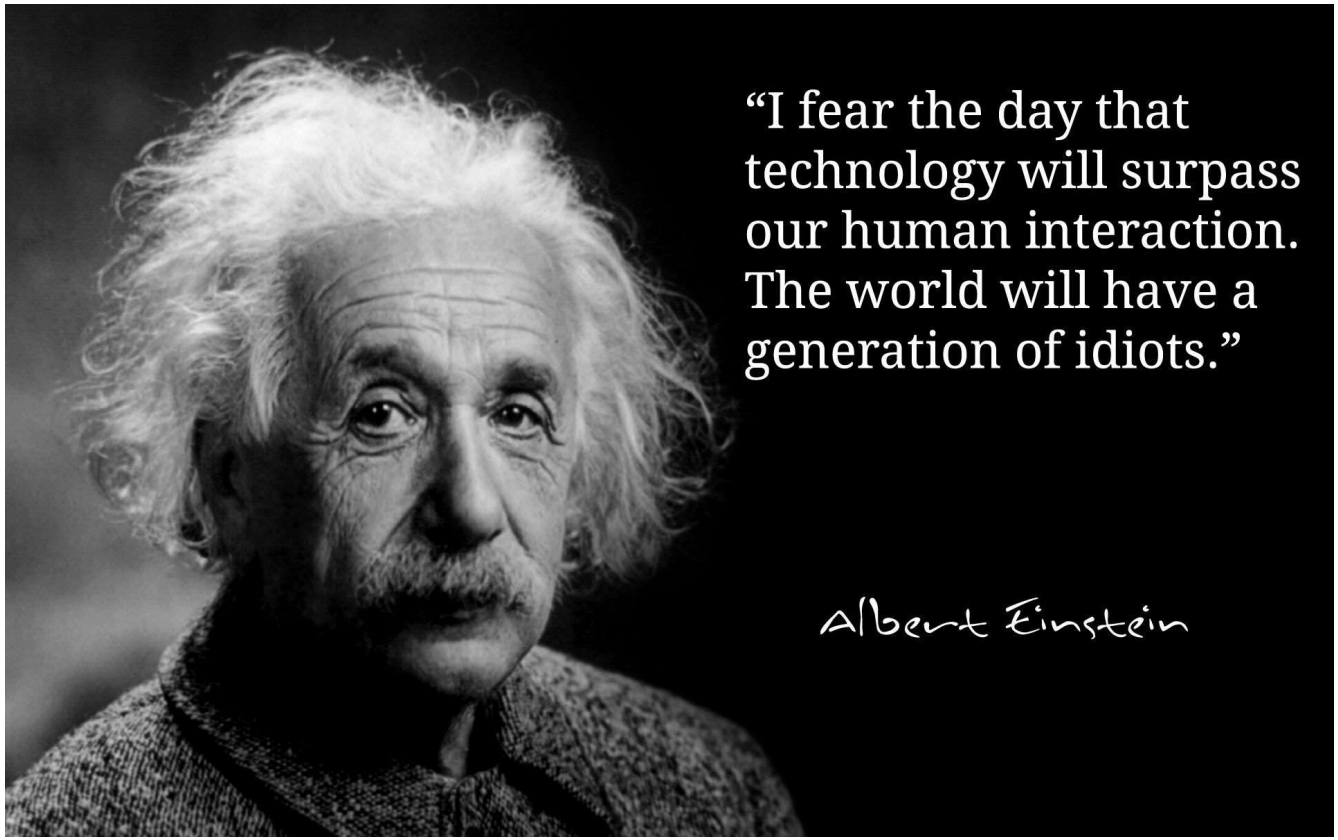
Conclusion:

A cybersecurity threat is a malicious and intentional attempt by a person or organization to enter another person's or organization's network without authorization in order to harm, damage, or steal IT assets, computer networks, intellectual property, or any other kind of sensitive data.

In cyber world, cyber threats can hack or damage your sensitive information and it has negative impact on individuals or organizations.

The attacks themselves are getting worse in these ways:

1. They’re getting more severe
2. They are getting more complex.
3. They’re getting more common.


“I fear the day that technology will surpass our human interaction. The world will have a generation of idiots.”
Albert Einstein

Thankyou